

November 20, 2020

If $B = (R, L)$ is a weighted bipartite graph with adjacency matrix M and having the following properties:

- $\forall u \in S, S \subseteq R, \deg(v) = |L|$
- $\forall u \in S, \forall v \in L, w((u, v)) = 1$

then $\text{per}(M) = \text{per}(M') \times \binom{|L| - |R| + |S|}{|S|}$, where M' is the adjacency matrix of the subgraph of B induced on all vertices adjacent to edges of weight ≥ 1 .

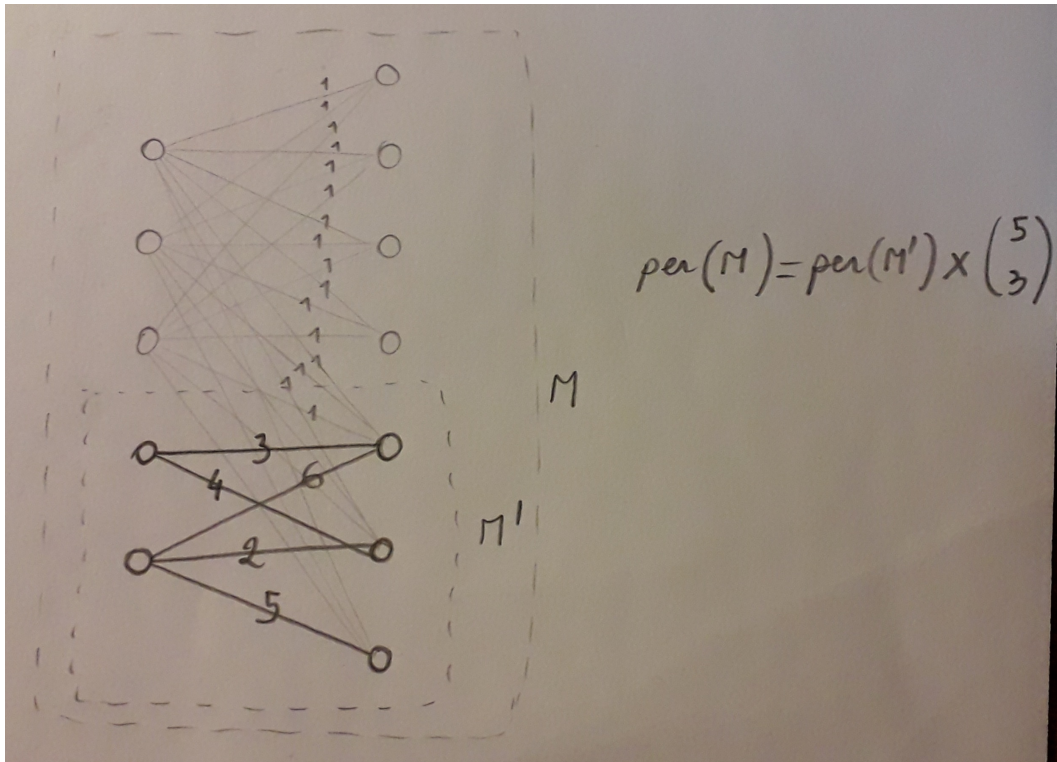


Figure 1: